CASAHERRERA
MANUFACTURERS OF FOOD MACHINERY

CHINS 103

CLEANING THE OVEN

CLEANING SHOULD BE DONE ONLY BY PERSONS WHO HAVE BEEN ESPECIALLY TRAINED TO CLEAN THIS MACHINE

CAUTION!
THE WORKING AREA WILL LIKELY BE WET AND SLIPPERY: BE CAREFUL! BE CERTAIN TO WEAR PROTECTIVE CLOTHING, INCLUDING BUT NOT LIMITED TO SAFETY GOGGLES, RUBBER GLOVES, RUBBER BOOTS, CHEMICAL-RESISTANT APRON, AND RESPIRATOR, ESPECIALLY WHEN USING SOLVENTS OF ANY KIND.

1. **ELECTRICAL SAFETY DISCONNECTS** should be turned to **"OFF".**
2. Do NOT get water on oven belts or electrical boxes! Cover electrical enclosures with plastic sheeting/
3. Do NOT adjust burners and other controls. Cleaning is NOT the time to make operator adjustments. Besides, the power should be OFF!
4. **ALL CLEANING SHOULD BE DONE AFTER THE OVEN HAS COOLED TO ROOM TEMPERATURE**.

EQUIPMENT/MATERIALS NEEDED:

- Wire brush
- Heavy-duty nylon scrubbing pad
- Putty knife (1" okay)
- Air Hose/Nozzle with low-pressure tip
- Vacuum cleaner
- Clean cotton cloths
- Safety goggles
- Mild detergent
- 5-gallon bucket
- Rubber gloves
- Rubber boots
- Padlock and safety lockout device

<u>DAILY PROCEDURE</u>: Wipe exterior surfaces with a clean damp cloth.

<u>WEEKLY PROCEDURE</u>

1. Remember to lock out all electrical power by turning the safety disconnect switch to "**OFF". USE OF A PERSONAL LOCKOUT DEVICE IS STRONGLY RECOMMENDED**!

2. Open all side doors and remove end panels as required

3. Vacuum interior of oven. While wearing goggles, blow out areas that cannot otherwise be reached

4. Using detergent solution, clean outside panels, inner insulating panels, hood and exhaust stack.

5. Oven belts: Loosen any burned-in deposits with wire brush. Empty catch-trays, if fitted. Check built-in conveyor cleaning brushes for wear and possible need of adjustment.

6. Replace cleaned outer panels, insulating inner panels, and guards. Report any damaged parts to the maintenance department.

7. Remove all items associated with cleaning from the food production area, storing them according to instructions from your supervisor.

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GENERAL OVEN MAINTENANCE

Other sections of this manual provide details on specific areas, such as bearings, drives, or belts.

NOTICE!
THE INFORMATION IN THIS SECTION IS FOR THE DESIGNATED PLANT MAINTENANCE PERSONNEL ONLY. NONE OF THESE THINGS SHOULD BE DONE BY THOSE WHO HAVE NOT BEEN PROPERLY TRAINED TO DO THE WORK!

WARNING!
THE ELECTRICAL POWER USED IN THESE MACHINES IS DANGEROUS! CONTACT WITH LIVE ELECTRICAL PARTS CAN RESULT IN SEVERE INJURY OR FATALITY! THE FOLLOWING PRECAUTIONS MUST BE TAKEN:

1. Power must be turned off when any maintenance or cleaning work is performed on the unit. Power should be turned off at the disconnect switch to prevent accidental starting of the equipment.

2. **BE SURE TO USE A SAFETY LOCKOUT DEVICE WHEN WORKING ON THIS EQUIPMENT**!

3. If power MUST be on, such as for maintenance troubleshooting, only qualified persons who are experienced and understand the hazards present should perform such work OR EVEN GO NEAR THE MACHINES. Work should be performed with at least TWO such trained persons present. Also, note that maintenance work is often performed at the same time that sanitation people are cleaning nearby machines. They may be using hose-directed water, spray bottles, damp rags, etc., and will set panels aside while cleaning certain areas. If an object is put on a conveyor belt that is then put into motion, severe damage to persons or property could occur!

4. We strongly encourage all persons connect with maintenance to be trained in emergency procedures, such as CPR.

5. Qualified maintenance electricians should check the system grounding periodically, to ensure its integrity.

INSPECT THE FOLLOWING ITEMS AT LEAST ONCE A WEEK:

MECHANICAL:

- Sprockets, chains, and belts - Look for signs of abnormal wear.
- Check alignment of sprockets with drive-chain.
- Check belt alignment with pulleys.
- Inspect belt for fraying, de-lamination, or other deterioration.
- Idlers should be keeping the belt or chain under correct tension.
- Check speed reducer gearbox weekly for leaks.
- Variable-Speed Pulleys/Drives should be cycled through their FAST/SLOW range as least once a week to prevent "sticking" at a usual setting.

OVEN VENTING SYSTEM

All joints should be mechanically sound, with stack/s properly supported. If equipped with exhaust blower, inspect motor, mounts, and that fan is securely on shaft. Inspect for and remove any obstructions in the system. Unless revised by a qualified person, the system should be as originally installed and approved.

LUBRICATION

See the section of this manual labeled 'LUBRICATION' for general location of lubrication points, recommended lubricants, and general procedures. For specific details see the 'MINS' sheets (Manufacturer's Information Sheets) in various sections of this manual.

ELECTRICAL

Test the integrity of connections by gently pulling on the wires. If strands of wire are sticking out from terminal connections, or are missing or broken due to poor stripping during installation, correct the condition immediately. Do NOT wait until later!
Note the condition of edges over which the wiring passes. If any burrs are found, remove them. BE THOROUGH! If insulation has been worn through but the wire strands are not damaged, restore the insulation with approved heat-shrinkable tubing. If the strands have been cut into or nicked, replace that section. If necessary to replace a length of wire, USE ONLY THE SAME COLOR, GAUGE, AND INSULATION TYPE AS THE ORIGINAL!
Remove all items that are not a working part of circuits, such as extra fuses, switch modules, etc. Carefully clean inside the enclosure with a vacuum hose if available.
Look for signs of moisture in the enclosures. If any is found, report the condition immediately to supervision.

ELECTRICAL PART REPLACEMENT CAUTION

Although it is possible to find functional equivalents for any part, this should not be done, except in emergencies. What is an emergency? Potential or actual lost production. Normal preventive maintenance is NOT AN EMERGENCY SITUATION, since it takes place during scheduled 'down-time'. Also, parts are usually drawn from regular stockroom inventories which are supposed to be kept up by orders that make allowance for hard-to-get items. Occasionally, an exact replacement part may be temporarily not available, and a substitute part might be pressed into service. Example: A DPDT relay goes bad. None are in stock, but there are 3PDT relays, with sockets, that have the same contact rating. Temporarily installing one and disregarding the extra contacts would be a logical way to "get going again". But if the substitute became permanent, and this was the usual way of doing things, the differences between the panel and the electrical diagram could become very confusing, to say the least! So as soon as possible, restore things to their original condition. This keeps machines uniform, and allows any qualified person to perform work without having to guess about anything.

FUSE REPLACEMENT CAUTION

TO REDUCE THE RISK OF FIRE OR ELECTRIC SHOCK, REPLACE ONLY WITH THE SAME TYPE AND RATING OF FUSE.

The voltage rating of a fuse is determined by how much energy it can safely handle when a short-circuit occurs. If the fuse is too light-weight for the energy level of the circuit, two things are possible; (a) The power can "arc-over" the burned-out segment, and power will continue to flow. (b) The fuse tube assembly could rupture and scatter flaming debris in all directions. That is why higher-voltage rated fuses of a given amperage are sometimes much larger in dimension.

"SLOW" fuses are NOT INTERCHANGEABLE with "FAST" fuses. Some circuits are easily damaged, and a "FAST" fuse will open the power connection before real damage is done. On the other hand, most motors draw extra-high current for a couple of seconds when they first start building up speed. This is normal, and "SLOW" fuses allow for that without opening up. A "FAST" fuse would have to be a lot higher amperage to get past that "hump". Then, when the motor settles down to its normal current draw, the only thing protecting the motor against gradual overheating would be the contactor overload relay (if it works!).

PILLOWBLOCK/FLANGE BEARING INSTALLATION

Make sure that the shaft, especially the end, is free of burrs, or has not been slightly "mushroomed" by someone's beating on it. Setscrews should be clear of the bore. Also, they often cut into the surface they are screwed against, raising the metal. Since bearing/shaft fits are fairly close, this is often enough to prevent removal of the bearing without galling the shaft. Then it will be IMPOSSIBLE to get apart! THIS IS ESPECIALLY TRUE OF STAINLESS STEEL SHAFTS! If at all possible, dress down such burrs BEFORE attempting dis-assembly. Use a fine-cut 12-inch flat mill file. When installing the replacement bearing, note that it should slide smoothly into position on the shaft without being too tight or loose. NEVER HAMMER THE ENDS OF THE INNER RACE SINCE THEY ARE NOT HARDENED STEEL. If necessary to apply force, GENTLY tap a brass bar or pipe against the INNER race to coax the bearing into place. If more force seems to be needed, DO NOT PROCEED until the cause is corrected!

Fasten the bearing housing to its mounting place on the machine. Rotate the shaft by hand to make certain that it turns freely. Then tighten the setscrews securely onto the shaft. Recommended minimum torque tightening values are shown in the table below.

TORQUE FOR TIGHTENING SETSCREWS			
		Minimum Recommended Torque	
Setscrew Diameter	Hex Size Across Flat	Inch/pounds	Foot/pounds
#10 (.190)	3/32	28	2.3
1/4	1/8	66	5.5
5/16	5/32	126	10.5
3/8	3/16	228	19.0

Oven Preventative Maintenance Schedule

ACTIVITY	DAILY	BI WEEKLY	WEEKLY	MONTHLY	Notes:
1. Visually inspect belt tracking	X				**Belt should track in center of rolls and touch guide rollers only occasionally.**
2. Lubricate oven bearings.		X			
3. Lubricate drive chains.		X			
4. Clean lenses on temperature sensors.				X	**Clean with soft cloth, as often as needed depending on environment**
5. Clean lens on flame detector and inspect spark gap on pilot burner.					**Same as above.**
6. Check gearbox oil				X	**Change yearly.**
7. Inspect belts and belt tension arms.			X		**Tension arms should not go below horizontal. Belts may need to be shortened.**
8. Inspect belt guide rollers and repair if necessary			X		

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OVEN BELT TENSION & ALIGNMENT PROCEDURE

The oven belts have been factory adjusted for the best possible running conditions, however, further adjustment maybe necessary after shipping or moving to a different location in the plant. Also, the belts will stretch after some use.

NOTE: THE FOLLOWING ADJUSTMENTS ARE BEST ACHIEVED WHEN OVEN IS HOT AND FULLY LOADED WITH PRODUCT.

I. PRELIMINARY INSPECTION

A. Insure that the drum bearings are lubricated and free to move along the guides.
B. Insure that the belt guide rollers are free. **DO NOT LUBRICATE**
C. Slide and lock the counter weights to the finest possible position wherein the belt drive is constant and not slipping. This adjustment is best achieved when oven is hot and fully loaded with product. (See Figure 1.)

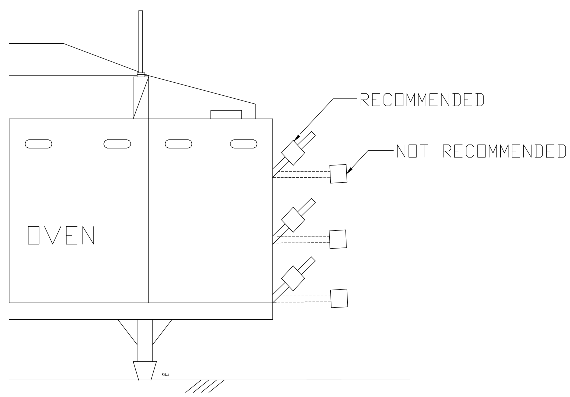

FIG. 1

D. Locate and lock take-up bearing to insure maximum take-up. (Figure 2.)

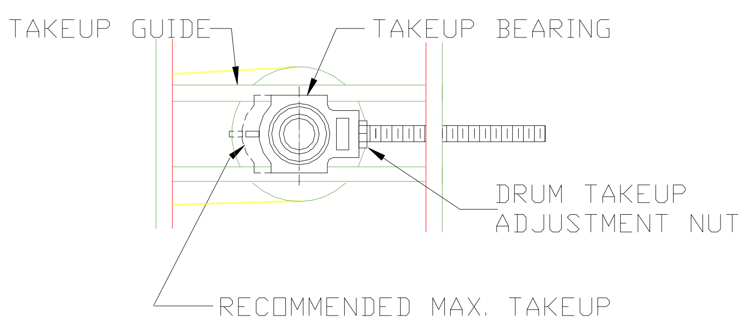

FIG. 2

II. BELT ALIGNMENT

A. When oven is hot and fully loaded with product, observe if the belt edges are continually making contact with the belt guide rollers on either side.

NOTE

> SOME LATERAL DRIFT (SKEW) OF THE BELT IS NORMAL, INCLUDING PERIODIC CONTACT OF THE BELT EDGES WITH THE GUIDE ROLLERS. ADJUSTMENTS IN THIS PROCEDURE ARE INTENDED TO REDUCE DRIFT AND CONTINUOUS GUIDE ROLLER CONTACT.

B. To reduce drift and continuous guide roller contact, adjust the drum bearing take-ups by turning one full turn and observing the result on the running belt for at least 3-5 minutes before making further adjustment.

C. Drift compensation is made by turning the bearing take-up nut on the side of drift counter clockwise. The same correction may be realized by turning the bearing take-up nut opposite the direction of drift clockwise. (See Figures 3A & 3B).

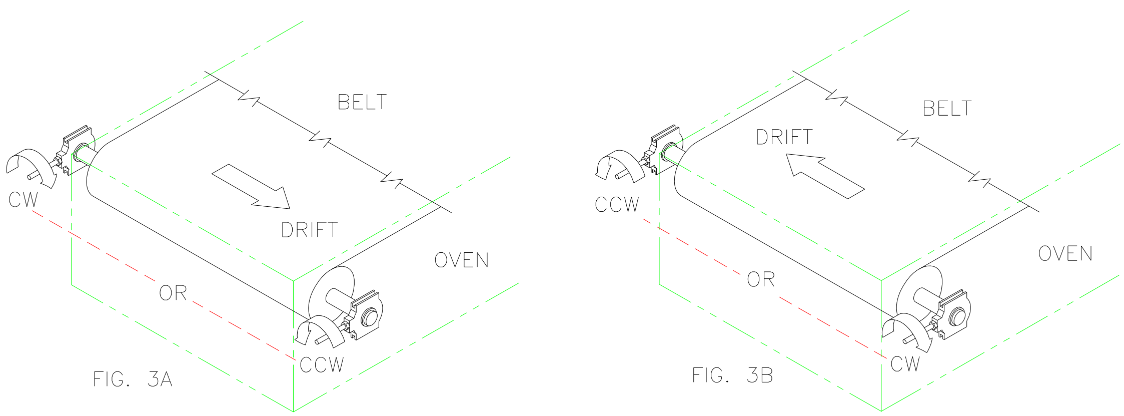

FIG. 3A FIG. 3B

NOTE:

DO NOT OVER-CORRECT. INITIAL BEARING TAKEUP SHOULD BE DONE IN ONE-TURN INCREMENTS; THEN PROGRESSIVELY SMALLER INCREMENTS AFTER A MINIMUM OF 2-5 MINUTE INTERVALS OF INSPECTION. IT IS NOT UNCOMMON FOR A BELT TO DRIFT CRITICALLY AFTER A 1 TO 2 HOUR PERIOD OF OPERATION. THEREFORE, CONSTANT INSPECTION IS RECOMMENDED.

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It is possible for a belt to be tracking properly at one end, but not the other. For example, at the exit end, the belt can be running centered on the roller, but at the input end the belt is over to one side. To correct this condition, follow the instructions below.

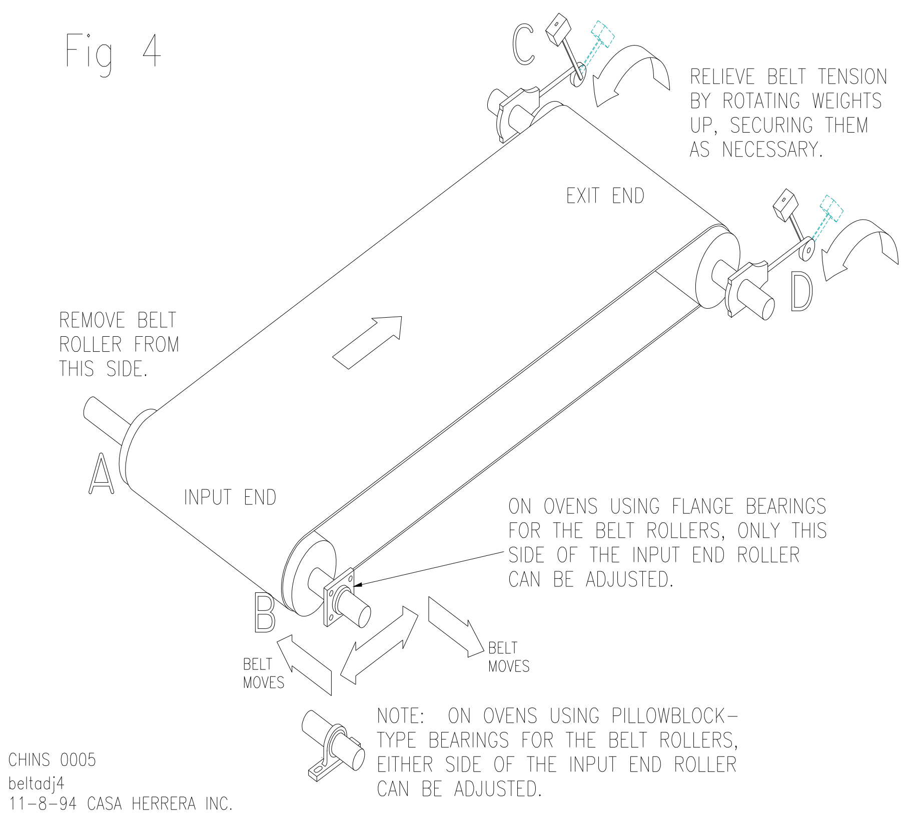
IMPORTANT! BEFORE ANY ADJUSTMENTS ARE DONE, THE OVEN MUST BE PROPERLY LEVELED AND SUPPORTED. OTHERWISE, THESE ADJUSTMENTS MAY HAVE LITTLE OR NO EFFECT.

PROCEDURE: (See Fig. 4)

1. Shut off the belt drive, and wait until it comes to a complete stop.
2. Disconnect power to the oven, and secure it using your personal lockout device.
3. Create some slack in the belt by rotating up the belt tension weights at the discharge end of the oven, securing them so that they cannot fall back into position and suddenly put tension on the belt. (See Fig. 4)

OVEN BAKE BELT TRACKING ADJUSTMENTS

Fig 4



FLANGE BEARINGS

<u>Left side flange bearing (‘A’) is NOT part of the tracking adjustment. DO NOT loosen it except to REMOVE the roller</u>. That side of the oven frame is cut out to allow removal of the belt roller.

4. At position ‘B’, (see diagram above) loosen the bolts holding the flange bearing.
5. a. To make the belt track more toward ‘B’, adjust the flange bearing toward the ‘EXIT’ end of the oven.
   b. To make the belt track more toward ‘A’, adjust the flange bearing toward the ‘INPUT’ end of the oven.

PILLOW BLOCK-TYPE BEARINGS

Either side of the roller may be used for adjustment.

IMPORTANT!! MAKE THE ADJUSTMENT IN SMALL AMOUNTS!!

6. Tighten the bearing mounting bolts.
7. After making sure that no tools are being left in the drive train area or bake belt, re-install the protective covers, and return the belt tension weights to their operating position.
8. Briefly run the bake belt to see the effect of the adjustment.

<u>INSTALLING AND ADJUSTING NEW BELTS</u>

NOTE: Two belt slide rail system is used on each belt. One system supports the top or product carrying portion and the other supports the lower or return portion.

Both types of belt slides should be adjusted so that the belt drops 1/8” to 1/4” between where it leaves the drum and contacts the slide.

DCO ovens have a slide rail system to guide the belt. The procedure for aligning the guides for the new belt is as follows:

1. Tie a nylon fishing line tightly around the front and rear drums as shown in figure 1. Make sure to do the adjusting procedure at A, B, C, D, E, and F as shown in figure 2.

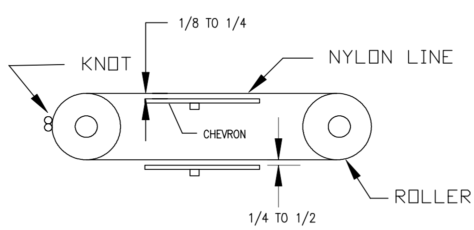


FIGURE 1

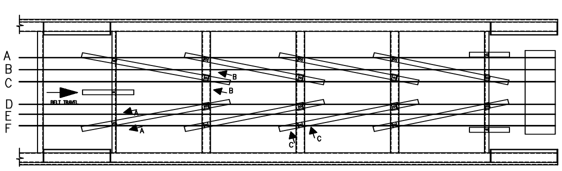


FIGURE 2

2. Adjust lower rails 1/8” to 1/4” below nylon lines. If lower than given figures add shims to the chevron bases as shown figure 3.

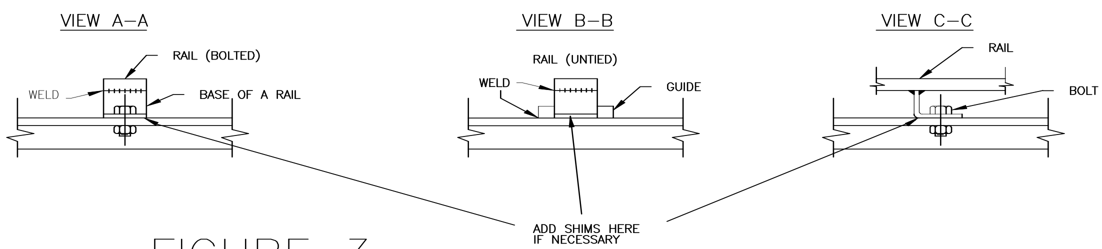


FIGURE 3

3. Insert “bubble” type level between rails and nylon lines and level rails across oven
4. Adjust upper rails 1/4” to 1/2” below nylon lines. If lower than given figures add shims to the chevron bases as shown in figure 3.
5. Insert “bubble” type level between rails and nylon lines and level rails across oven.

PARALLELING DRUMS OR ROLLERS

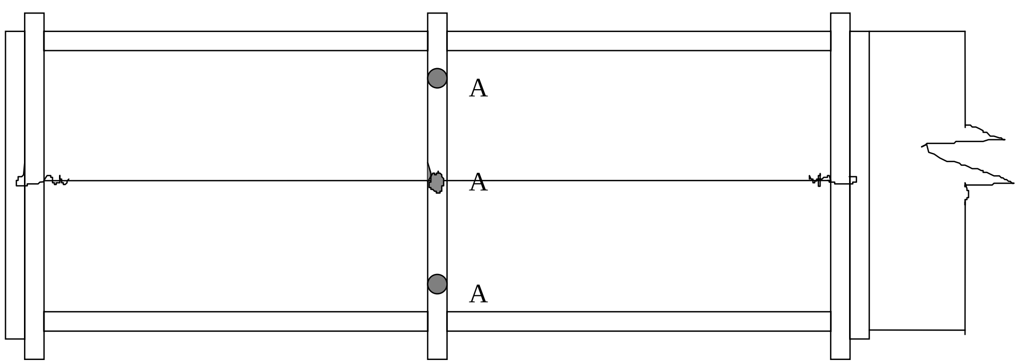


Figure 5

a. Clamp a third metal straight edge or flat bar across the oven roughly midway between the two drums, and mark where the centerline of the oven intersects the approximate centerline of the straightedge or flat bar (see figure 5).
b. Using a center finder such as the Racine “Mercury Balanced Miracle Point” mark each end of each drum at “top dead center” and 1/2” in from each end. Check that distance from mark to centerline of oven is the same for each side of drum.

Installation and Operating Recommendations

CB5 BAKING BANDS

The manufacturer guarantees quality of workmanship and the metal used in the fabrication of the Band to be as specified. Belts are tested for true travel and rigidly inspected before shipment. The care used in the installation of the new belt and the observance of good operating practices is insurance of long life and low maintenance costs. The following suggestions are offered for your guidance:

GENERAL Friction driven Woven Wire Belts are designed for operation over straight faced terminal pulleys. Flanged pulleys will not control side travel and are not recommended, nor should crowned pulleys be used. Terminal pulleys and snub rolls should be adjusted to be level and square to the conveyor centerline. To correct belt side travel re-check terminal alignment and make corrections to the belt supports as necessary. Permanent damage to the belt can occur if the terminals are "cocked". See Bulletin IC75 for further information on belt tracking. Supporting rolls must be level and at right angles to the conveyor centerline. Belt supporting surfaces should be checked for freedom from obstructions such as warped hearth plates or protruding framework, which might cause belt impingement. Check the belt take-up mechanism to make certain it is functioning properly and remains square to the conveyor centerline throughout its travel.

TO MAKE CB5 BAKING BANDS ENDLESS (See Figure 1)

For ease of handling, most belts, because of their length and weight, are shipped in two or more sections. Normal care should be used in uncrating to prevent damage to the mesh. The same method of joining sections applies to making the belt endless. Our practice is standard whether the belt is shipped in one continuous length or is made up of two or more sections. All belt lengths terminate with loose spirals. The last, loose spiral is left hand (counter clockwise winding). The leading, fixed Spited that is welded to the crimp rod at the opposite end, is right hand (clockwise winding). Extra crimped connectors and loose spirals are included with every belt shipped. The spirals have a long end loop to facilitate fixation and the crimped connector wires are longer than required for the belt width. **TO MAKE ENDLESS**-bring the two sections together with the edges of the belt in line. Mesh the end spirals together to permit the insertion of the five crimped connector wires. Make certain the end spirals so joined are of opposite weave-one right, the other left hand. After inserting the five connector wires, cut the crimped connectors to the proper length and weld the protruding ends of the connectors to the proper spirals. Caution: Before welding see that all spirals bottom or seat on the connector and lie flat.

DIRECTION OF BELT TRAVEL (See Figure 1) When installing the belt make sure the spiral LEADS the crimped rod to which it is welded at both ends.

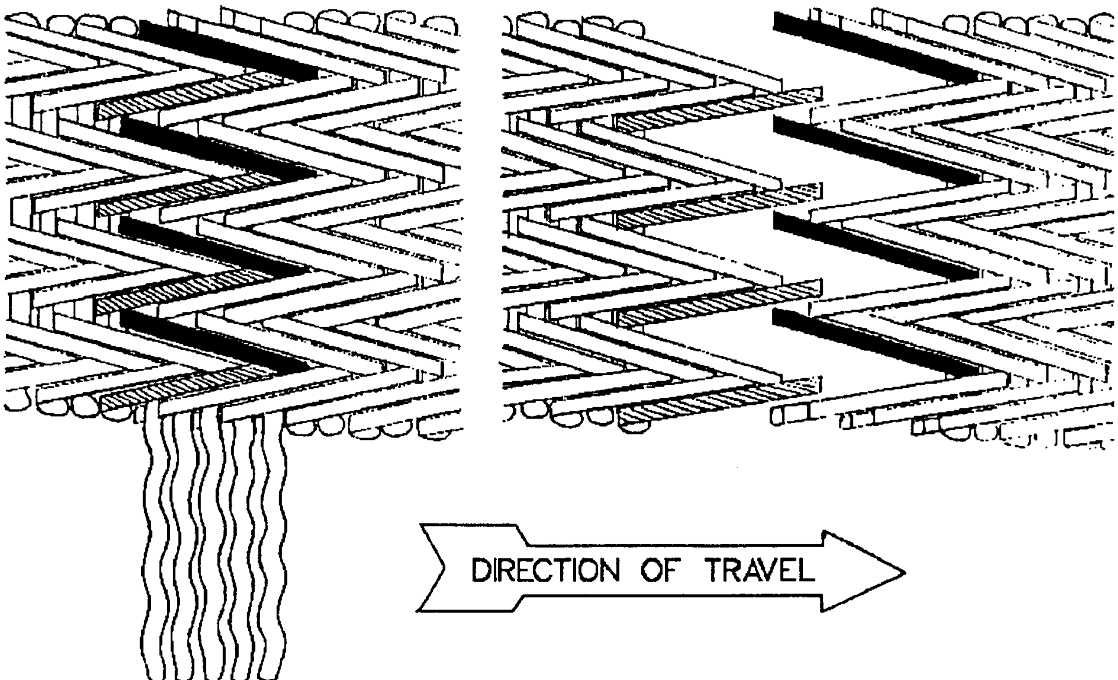


FIGURE 1

TO REMOVE STRETCH FROM CB5 BAKING BANDS

Normally, maximum stretch takes place during the initial period of operation. When stretch exceeds that which can be taken up by the TAKE-UP device, a section of belt must be removed. Tag the section to be removed, take out as much as possible without making the re-assembly too difficult. Be sure the first spiral cut at each end of the section to be removed is of the same turning. On one side cut the crimped wires, on the opposite side cut the spirals to which the crimp wires are welded (cut the spiral as close as possible to the weld to allow enough wire to re-weld to the new crimp). Then pull out the five crimped connector wires at each end of the section to be removed. The belt is now ready to be made endless again. Bring the two sections of the belt together with the edges in line. Mesh the end spirals together and insert the five crimped connector wires. Make certain the end spirals so joined are of opposite weave, one right hand and the other left hand. Cut the crimped connector wires to the proper length and weld the protruding ends to the proper spirals.

SUGGESTED ASHWORTH CB5 BAKING BAND CLEANING PROCEDURES

The accumulation of material in the mesh of a CB5 Band can be very detrimental to its useful life if allowed to progress too far. This material becomes very hard and causes spiral distortion, and eventual failure through fatigue, when the band flexes over the terminal drums, snub rolls, etc. At an advanced stage, the band is almost a solid object. The removal of this material at this point is extremely difficult; therefore, it is better to periodically clean the band to prevent this massive buildup.

If a significant buildup has occurred, the object of the cleaning process is to raise the temperature to the point where the accumulation will carbonize. It then can be more easily broken up and will fall out of the mesh. If possible, the oven temperature should be raised to about 900°F. with the band in motion. If the equipment is not capable of this level, add several extra burners at a convenient location so that at least a local section of the band will reach this temperature as it passes through. It is believed that the application of liberal amounts of vegetable oil to the band during the burn out procedure helps to dislodge the material by penetrating the mesh.

The risk of fire must be evaluated prior to this procedure. For instance, the oven should not contain an excessive amount of product debris that might ignite at the elevated temperature. Limit the amount of oil subjected to high temperatures as it may also ignite. Common sense and safety precautions must be followed.

A driven stiff rotary brush on one or both sides of the band will aid in extracting the particles, some people have also employed a “belt beater” roll to help break up and dislodge the carbon. No sharp corners or high pressure against the mesh surface should be used. Four pipe sections arranged in a square pattern and driven by a central shaft can simulate a square roll. It is preferred that these be made free to rotate to minimize any scuffing action on the mesh. The round surfaces of the pipe at the “corners” will not abuse the belt if properly used. A beater is not necessary or suggested for ordinary cleaning.

If steam cleaning is attempted, do not apply water or steam to a hot band as it may cause irreparable local distortion. The band (and oven) should be only at a temperature that will insure complete drying. Apply a coating of vegetable oil after the cleaning procedure to protect the steel band against rusting.

Cleaning effectiveness is limited in some cases, depending upon the particular degree and type of band contamination. Rather than having to resort to severe methods, it is better practice to include a cleaning procedure in the maintenance routine at prescribed intervals based upon the amount of goods baked that contain a high content of sugar and shortening. A mild accumulation should be easily removed by temperature and brushing alone.

These procedures carry no guarantee of success but they represent the accumulation of knowledge of various cleaning methods attempted by a number of users ASHWORTH, as a manufacturer of Baking Bands, does not have the opportunity to experiment with cleaning procedures.